Glomerular Disease — Nephritic Syndrome

Pathology GUE Block · MEDucation by Mattin Majid

Part II of this LG — continued from Part I: nephrotic syndrome.

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the three types of RPGN by immunologic findings, and which immunofluorescence pattern (linear/granular/scanty) corresponds to each?
2. What EM finding is pathognomonic of post-streptococcal GN, and where is it located relative to the GBM?
3. Which gene/collagen chain is defective in Alport syndrome, and what is the classic EM GBM appearance?

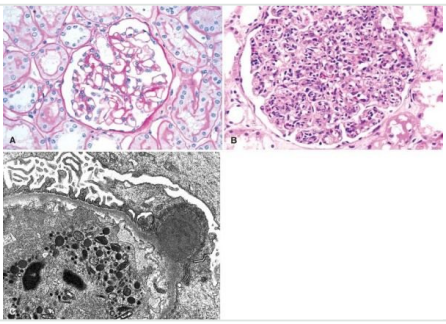
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Nephritic syndrome — definition

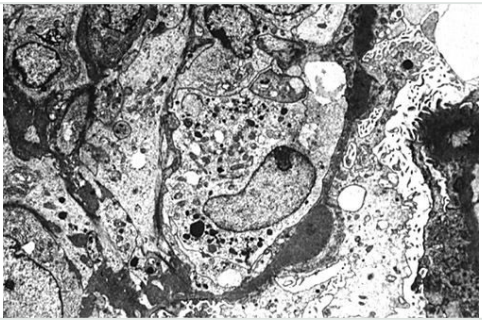
What	Characterized by **inflammation in the glomeruli.**
Clinical findings	**Hematuria** (red blood cells & red cell casts in urine); proteinuria usually **subnephrotic**, with or without edema; **oliguria**; **azotemia**; **hypertension**.

2 Post-streptococcal GN (Acute Proliferative GN, APGN)

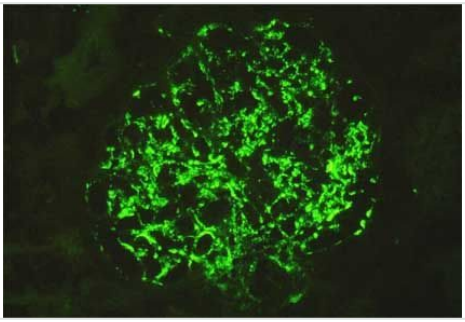
WHO	**1 to 4 weeks** after a streptococcal infection of pharynx or skin (impetigo). Mostly in **children (6–10 years)**, but adults can be affected.
Etiology & pathogenesis	**Group A β-hemolytic streptococci** — certain strains (1, 4, 12) are nephritogenic. Increased titers of antistreptococcal antibodies — anti-streptolysin O (ASO), anti-DNAase B, anti-cationic proteinase. **Granular immune complex deposits** in glomeruli.
Morphology	Glomerular hypercellularity is due to **intracapillary leukocytes and proliferation of intrinsic glomerular cells**. EM: typical electron-dense **subepithelial "hump"** (above GBM and below the parietal epithelial cell layer), and a neutrophil in the capillary lumen.
Immunofluorescence	**Granular** immune complex deposits (IgG, IgM, and C3) in mesangium and glomerular BM.



A: normal glomerulus. B: glomerular hypercellularity (intracapillary leukocytes + proliferation of intrinsic cells)



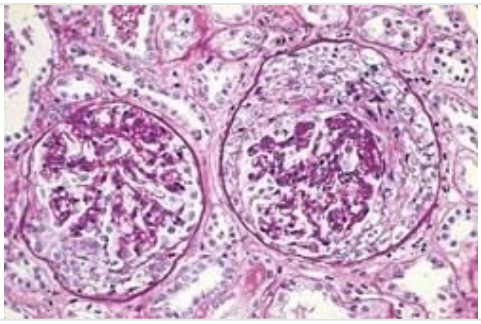
EM — typical electron-dense subepithelial "hump" and a neutrophil in the capillary lumen



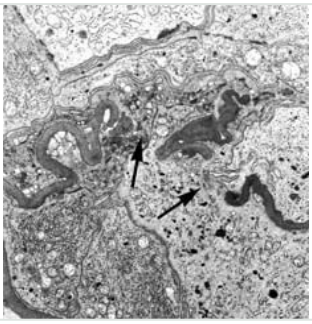
Granular immune complex deposits (IgG, IgM, C3) in mesangium and GBM

3 Rapidly Progressive (Crescentic) GN

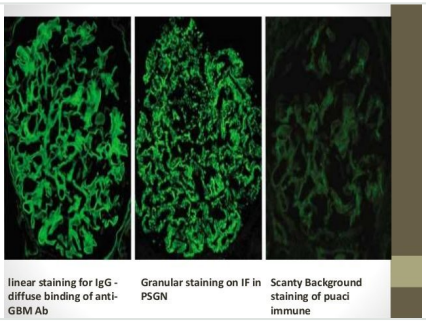
What	Rapid and progressive loss of renal function with severe oliguria. If untreated: death from renal failure within weeks to months.
Crescents	Produced by: proliferation of **parietal epithelial cells**, fibrin deposition, and infiltration of leukocytes.
Classification (by immunologic findings)	**Type I RPGN** (anti-GBM antibody) — **Goodpasture syndrome**; shows **linear** IF staining for IgG (diffuse binding of anti-GBM Ab). **Type II RPGN** (immune complex) — postinfectious; shows **granular** IF staining (as in PSGN). **Type III RPGN** (pauci-immune) — lacks anti-GBM antibodies or immune complexes; **ANCA-associated**; scanty background IF staining.
Morphology	PAS: collapsed glomerular tufts and the crescent-shaped mass of proliferating cells and leukocytes internal to the Bowman capsule. EM: characteristic wrinkling of GBM with focal disruptions in its continuity.



Crescentic GN (PAS) — collapsed glomerular tufts and crescent-shaped mass of proliferating cells/leukocytes internal to Bowman capsule



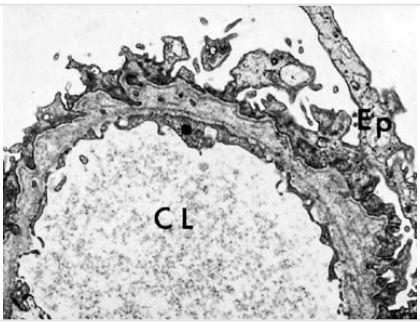
EM — wrinkling of GBM with focal disruptions in its continuity (arrows)



Immunofluorescence by RPGN type: linear IgG (Goodpasture) · granular (post-infectious/PSGN) · scanty background (pauci-immune)

4 Hereditary nephritis — Alport syndrome

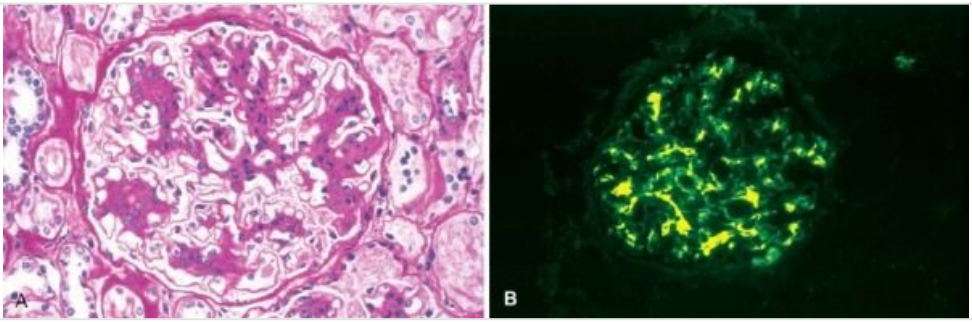
Inheritance	**X-linked form:** most common; males express full syndrome, females are carriers. **AD and AR forms:** rare; both males and females express full syndrome.
Clinical	When fully developed: nephritis progressing to CRF, nerve deafness, and eye disorders (lens dislocation). In carriers: typically limited to hematuria.
Pathogenesis	Defective GBM synthesis because of mutation in the gene for the α5 chain of **type IV collagen.**
Morphology	EM: GBM shows irregular foci of thickening alternating with attenuation (thinning), with a **"basket-weave" appearance.** Similar alterations can be found in tubular BM.



Alport syndrome — EM "basket-weave" GBM appearance

5 IgA Nephropathy (Berger disease)

WHO	**Most common type of GN worldwide.** Can affect all ages but most common in the **2nd and 3rd decades** of life.
What	Presence of prominent **IgA deposits in the mesangium**. Can be suspected by light microscopy, but diagnosis is made **only by immunofluorescence.**
Clinically	Frequent cause of recurrent gross or microscopic hematuria. Mild proteinuria usually present. Occasionally develops nephrotic syndrome. Rarely develops rapidly progressive crescentic GN.
Pathogenesis	Genetic or acquired abnormality of immune regulation leading to increased mucosal IgA synthesis in response to respiratory or gastrointestinal exposure to environmental agents (e.g. viruses, bacteria, food proteins). IgA-containing immune complexes are then trapped in the mesangium.
Morphology	Light microscopy: mesangial proliferation and increased mesangial matrix. Immunofluorescence: deposition of IgA, principally in mesangium.



A: LM — mesangial proliferation and increased mesangial matrix. B: IF — IgA deposition, principally mesangial

P Must-remember pearls

- Nephritic syndrome triad/pentad: **hematuria (RBC casts), subnephrotic proteinuria, oliguria, azotemia, hypertension.**
- APGN: subepithelial **"humps"** on EM, **granular** IF deposits (IgG/IgM/C3); follows strep throat/impetigo by 1–4 weeks, mainly children 6–10y.
- RPGN by IF pattern: **Type I linear** (anti-GBM, Goodpasture) · **Type II granular** (immune complex, postinfectious) · **Type III pauci-immune/scanty** (ANCA-associated).
- Crescents = proliferating **parietal epithelial cells** + fibrin + leukocytes, internal to Bowman capsule.
- Alport syndrome: **X-linked** most common (males affected, females carriers); defective **α5 chain of type IV collagen**; EM shows **"basket-weave" GBM**; triad = nephritis + deafness + lens dislocation.
- IgA nephropathy (Berger disease) is the **most common GN worldwide**; diagnosis requires **immunofluorescence** showing mesangial IgA deposits; classic presentation is recurrent gross hematuria, often with a preceding mucosal infection.

Q Test yourself

1. A 7-year-old develops hematuria and hypertension 2 weeks after a sore throat. Renal biopsy EM shows a subepithelial electron-dense hump. What is the diagnosis?

A. Minimal change disease
B. Post-streptococcal GN
C. IgA nephropathy
D. Alport syndrome

2. A patient with rapidly progressive renal failure and hemoptysis undergoes renal biopsy showing crescents, with immunofluorescence showing linear staining for IgG along the GBM. What is the diagnosis?

A. Type I RPGN (Goodpasture syndrome)
B. Type II RPGN (immune complex)
C. Type III RPGN (pauci-immune)
D. Post-streptococcal GN

3. Crescents in rapidly progressive glomerulonephritis are composed of which combination of elements?

A. Podocytes, mesangial cells, and eosinophils
B. Proliferating parietal epithelial cells, fibrin, and leukocytes
C. Endothelial cells and IgA deposits
D. Tubular epithelial cells and amyloid

4. A young man presents with hematuria, sensorineural hearing loss, and lens dislocation. His mother has isolated hematuria. Renal biopsy EM shows a basket-weave GBM pattern. Which gene/protein defect is responsible?

A. WT1
B. Alpha-5 chain of type IV collagen
C. PKD1
D. Podocin

5. A 22-year-old develops gross hematuria 1-2 days after an upper respiratory infection. Light microscopy shows mesangial proliferation; the diagnosis is confirmed by which test showing mesangial deposits?

A. Congo red stain
B. Electron microscopy alone
C. Immunofluorescence for IgA
D. PAS stain alone

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · B — A subepithelial 'hump' on EM, with granular IF deposits and a preceding strep infection, defines post-streptococcal (acute proliferative) GN.

2 · A — Linear IgG staining along the GBM with crescents and pulmonary hemorrhage describes Type I RPGN / Goodpasture syndrome (anti-GBM antibody disease).

3 · B — Crescents form from proliferation of parietal epithelial cells, fibrin deposition, and infiltrating leukocytes.

4 · B — Alport syndrome is caused by mutation in the gene for the alpha-5 chain of type IV collagen, producing a basket-weave GBM on EM.

5 · C — IgA nephropathy can be suspected on light microscopy but is diagnosed only by immunofluorescence showing mesangial IgA deposits.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam